

VAYUSANGAM | OpenAQ COLLECTION STATUS

Status addendum to the dataset prerequisite inventory | Checked 30 Sep 2026, 11:47 IST

FULL HISTORICAL PULL IS IN PROGRESS - NOT COMPLETE

The raw cache is growing, while the completed manifest and normalized CSV still describe the validated 3-day sample. This page supersedes the earlier OpenAQ counts on the inventory pages that follow.

Verified sample currently in the normalized files

SAMPLE WINDOW	STATION	OBSERVED VALUES
72 hourly rows 2026-09-28 to 2026-09-30	1 station GK1 (Oberoi Terrace)	PM2.5: 51 Temperature: 51 Relative humidity: 51

Other requested variables have zero normalized observations in this sample; nulls are preserved, not filled. Sample validation passed. This is an API sample, not scientific validation.

Collection inventory at this check

Artifact / evidence	Current verified state
Station discovery	220 locations; 209 report a PM2.5 sensor; 167 have declared PM2.5 history overlapping 2024-01-01 to 2026-09-30.
Sensor metadata	9 sensor lookups returned persistent HTTP 500 and are marked error; availability for those locations is unknown.
Hourly output / manifest	72 rows for 1 station. Manifest still has sample_mode=true and dates 2026-09-28 to 2026-09-30.
Raw hourly cache	3,941 JSON response files at check time. This includes previous cache plus partial full-range responses; it does not prove complete coverage.
Quality report	23 sample quality records; validator passed for the sample only.

Full-range collection and next steps

The configured source period remains 2024-01-01 through 2026-09-30. Raw monthly chunks are being cached resumably, but no complete full-range manifest, hourly CSV, or quality report exists yet. If collection stops, rerun `.venv/bin/python scripts/data_collection/openaq/download_openaq.py`; cached responses are reused. After it completes, run `.venv/bin/python scripts/data_collection/openaq/process_openaq.py` and `.venv/bin/python scripts/data_collection/openaq/validate_openaq.py`.

Other inventory conclusions on the following pages remain unchanged: historical surface meteorology is present; forecast NWP, PBLH/upper-air data, CAMS and FIRMS remain absent or unverified. OpenAQ hourly values are not a verified CPCB feed.

VAYUSANGAM | DATASET & PREREQUISITE STATUS

Repository data inventory against the supplied SIH blueprint requirements

Rechecked: 30 Sep 2026 | Scope: current files under data/

AT A GLANCE

Downloaded: OpenAQ station/hour data (large raw cache; normalized run is explicitly one-station sample), Open-Meteo historical reanalysis for 10 NCR/outer-ring points, OpenStreetMap context layers, derived feature CSVs.

Not found locally: CAMS, NASA FIRMS fire detections, direct Copernicus CDS ERA5 files, archived forecast NWP fields.

Key modeling gap: meteorology has surface variables only. PBLH, upper-air temperatures (T925/T850), radiation and forecast fields are not present in the meteorology manifest or processed CSV.

DETAILED INVENTORY

STATUS | DATASET / BLUEPRINT USE | LOCAL EVIDENCE | WHAT IS ACTUALLY PRESENT
PARTIAL | CPCB hourly air-quality observations, NCR, 2019-2025; PM2.5, PM10, NO2/NOx, O3 | data/air_quality/air_quality_hourly.csv; raw/hours/*.json; station metadata, availability and quality reports; collection_manifest.json | Source is OpenAQ, not a verified CPCB feed. Raw cache has 3,165 hourly JSON files, 211 sensor JSON files and 220 listed station records per dictionary. However collection_manifest.json says sample_mode=true and downloaded_station_ids contains just 1 station (2860223), requested 2026-09-28..30. The normalized CSV and raw cache do not prove a complete NCR 2019-2025 training set. Parameter availability varies; check quality report before relying on each pollutant.

PRESENT (PROTOTYPE) | Historical meteorology for feature/training inputs | data/meteorology/raw/*.json; processed/meteorology_hourly.csv; meteorology_manifest.json and DATA_DICTIONARY.md | Open-Meteo historical archive, ERA5/ERA5-Land-derived. Ten selected NCR/outer-ring points; 120,480 hourly rows per dictionary covering 2024-01-01 to 2026-09-30. Surface temperature, RH, pressure, 10m wind, cloud and precipitation. This is reanalysis-derived history, not an archived operational forecast.

MISSING | Forecast NWP archive (GFS/Open-Meteo forecast or NCUM) | No forecast archive/manifest located under data/ | Required forecast input for operational 72-hour inference and forecast-input skill evaluation.

MISSING | PBLH, pressure-level temperature (T925/T850), radiation / vertical meteorology | Not listed in meteorology manifest or processed CSV (13 columns documented) | Needed for inversion strength, ventilation coefficient and aerosol-feedback features. Wind is 10m surface wind; no PBL-mean wind evidenced.

MISSING | CAMS composition forecast, PM/AOD baseline/features | data/cams contains no data files beyond .DS_Store | No downloaded CAMS data found.

MISSING | NASA FIRMS fire detections with FRP/location/time | data/fires contains no data files beyond .DS_Store | No downloaded active-fire data found. No evidence here of a configured MAP_KEY.

MISSING LOCALLY | Direct Copernicus CDS ERA5/ERA5-Land | data/era5 contains no data files beyond .DS_Store | The Open-Meteo archive is already derived from ERA5/ERA5-Land, but it is not a direct CDS download. Dictionary says direct CDS request is blocked on ~/.cdsapirc. Do not count this as independent additional data.

PRESENT (LIMITED) | Geospatial/static covariates | geospatial/raw/osm_roads.json, osm_landuse.json, osm_boundaries.json; geospatial_manifest.json | OSM roads, landuse tags and boundary geometry downloaded. Roads/landuse have no geometry, so cannot provide distance or point-in-polygon features. Delhi district boundaries are absent; district join is incomplete. Elevation/building density layers are not evidenced.

PRESENT | Derived feature tables | features/air_quality_features.csv; features/meteorology_features.csv | Feature products exist, but do not fill missing upstream observations, upper-air/forecast weather, CAMS or FIRMS. Validate coverage and provenance before evaluation.

PREREQUISITE / ACCESS READINESS

PARTIAL: OpenAQ API collection has been run; latest collection manifest explicit

ly labels its normalized run as a one-station sample. Historical raw cache is broader but is not evidence that the target CPCB training dataset is complete.
NOT VERIFIED: CPCB/data.gov.in access, licences/keys, FIRMS MAP_KEY, CDS account credentials, and NCUM participant access. These are account/access prerequisites, not downloaded datasets; local file inspection cannot prove account status.
PRESENT: OSM source files and manifest; manifest records ODbL attribution.
PARTIAL: Open-Meteo archive data present. It does not satisfy forecast archive or upper-air/PBLH requirements.

READINESS CONCLUSION

Current data/ supports a prototype using OpenAQ observations, surface historical weather, derived features and basic OSM context. It is not yet sufficient to substantiate the full training/evaluation dataset or the required coupled forecast inputs. Highest-priority data gaps are: (1) verified multi-year CPCB/OpenAQ station history with intended date coverage, (2) forecast NWP archive, (3) PBLH and pressure-level/radiation meteorology, and (4) CAMS and FIRMS products.

METHOD / LIMITS

Statuses are based on the current data/ files, manifests and data dictionaries, checked 30 Sep 2026, plus the prerequisite list supplied with the request. A file's presence does not establish scientific completeness, quality, legal eligibility or training readiness. Empty source folders containing only .DS_Store are marked missing. Blueprint PDF checked: SIH26082_VayuSangam_Blueprint.pdf (28 Sep 2026); its text was not extractable in this environment, so exact line wording is cross-referenced to the supplied detailed requirement list.